

Quality Assurance of a Diagnostic CT Machine

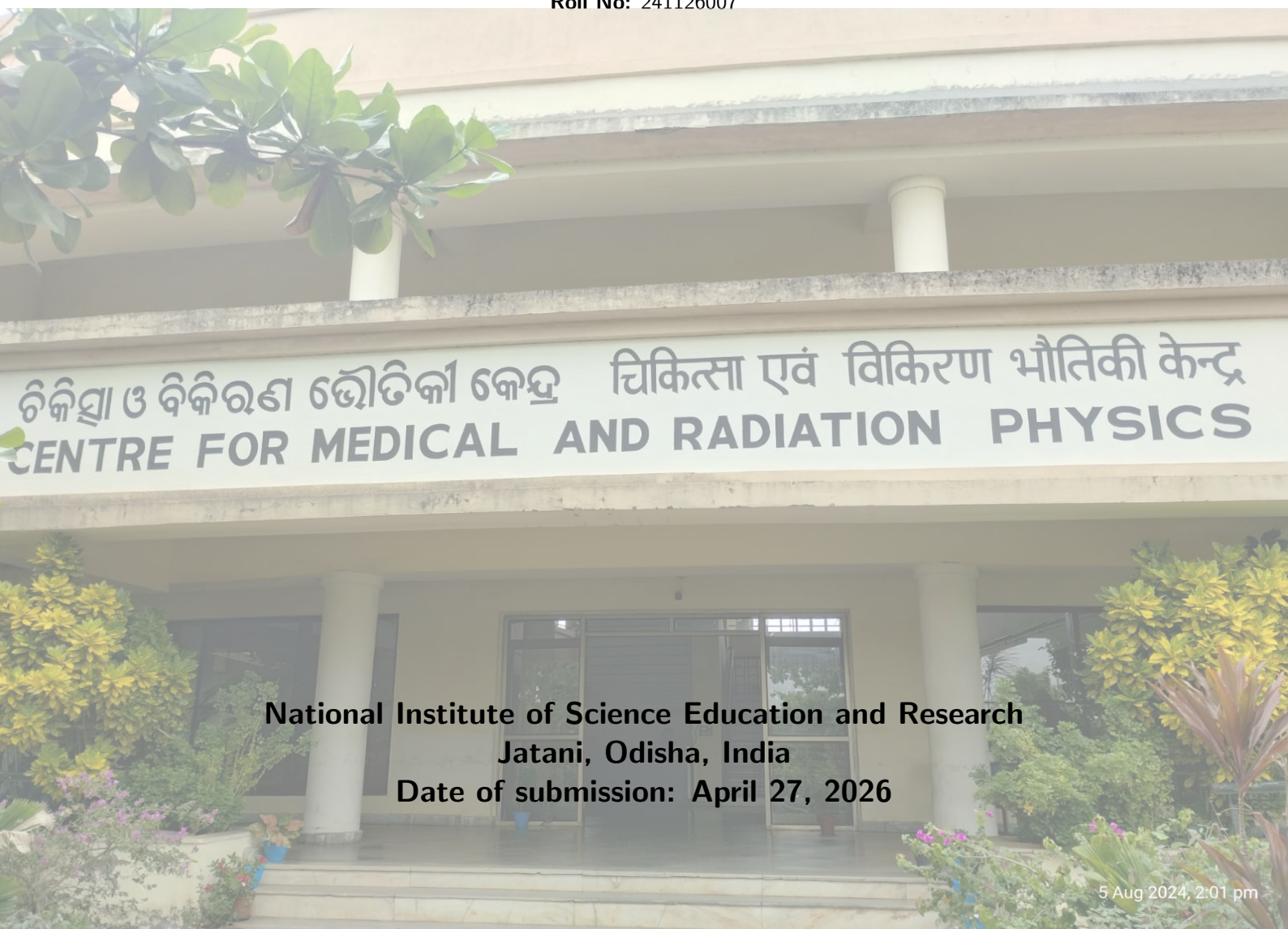


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Contents

1	Objective	1
2	Apparatus	1
3	Theory	1
3.1	Introduction to Computed Tomography Quality Assurance	1
3.2	Basic Principles of Computed Tomography (CT)	1
3.3	Description of the CT Machine Components	1
3.3.1	The Scanning Gantry	2
3.3.2	The Examination Table (Couch)	2
3.3.3	The Computer and Reconstruction System	2
3.3.4	The Control Console	2
3.4	Facility Design and CT Room Layout (AERB Guidelines)	3
3.5	Quality Assurance (QA) Parameters	4
3.5.1	Mechanical and Geometric Tests	4
3.5.2	Tests for High-Frequency Generators	4
3.5.3	Image Quality Evaluation	4
3.5.4	CT Dosimetry and Radiation Safety	5
3.6	Quality Assurance Equipment and Instrumentation	5
3.6.1	GE Optima CT580W CT Scanner	5
3.6.2	RTI Piranha Multi-Meter	6
3.6.3	RTI CT Ionization Chamber (10 cm)	6
3.6.4	Pro-CT MINI Phantom	6
4	Observation	7
4.1	Mechanical Tests	7
4.1.1	Gantry Tilt: Accuracy of the tilt Indicator	7
4.2	Radiation Safety Test of Computed Tomography Equip- ment	8
4.2.1	Slice Thickness	8
4.2.2	Accuracy of Operating Potential	9
4.2.3	Accuracy of Timer	9
4.2.4	Linearity of Radiation Output	9
4.2.5	Reproducibility of Radiation Output	9
4.2.6	Radiation Dose Test Measurement of Weighted CT Dose Index ($CTDI_w$)	9
4.2.7	Low Contrast Resolution	10
4.2.8	High Contrast Resolution	10
4.2.9	Radiation Leakage Level from X-ray Tube Housing	10
4.2.10	Radiation Protection Survey of the Installation	11
5	Conclusion	11

1 Objective

To comprehensively evaluate the mechanical and dosimetric integrity of a diagnostic Computed Tomography (CT) scanner in accordance with established Quality Assurance (QA) protocols.

2 Apparatus

- Diagnostic CT Scanner
- Standardized CT QA Phantoms (Water, Slab, Catphan or equivalent)
- Positioning lasers and metric measuring tools

3 Theory

3.1 Introduction to Computed Tomography Quality Assurance

A Quality Assurance (QA) program in diagnostic radiology is a comprehensive set of protocols and procedures designed to ensure that diagnostic images are of sufficiently high quality to provide adequate diagnostic information, achieved at the lowest possible cost and with the least possible radiation exposure to the patient [1].

For a Computed Tomography (CT) unit, QA protocols must strictly address image quality, geometric alignment, and dosimetric output. Furthermore, when CT images are utilized for radiotherapy treatment planning (CT-Simulation), the spatial integrity of the scanner and the consistency of electron density values across the CT, Treatment Planning System (TPS), and Digitally Reconstructed Radiographs (DRRs) must be rigorously verified [2].

Quality assurance testing guarantees two fundamental tenets of radiation protection:

1. The radiation dose received by the patient for diagnostic purposes is optimally minimized (ALARA principle).
2. High image quality facilitates accurate diagnosis, thereby scientifically justifying the radiation exposure.

3.2 Basic Principles of Computed Tomography (CT)

Computed Tomography (CT) is a highly sophisticated medical imaging modality that utilizes X-ray radiation to produce cross-sectional images (slices) of the human body. Unlike conventional projection radiography, where a three-dimensional anatomical volume is superimposed onto a two-dimensional plane, CT generates true cross-sectional maps of X-ray attenuation coefficients. It accomplishes this via computationally reconstructing multiple angular projections acquired as an X-ray tube and detector array rotate around the patient.

The fundamental principle revolves around the measurement of radiation attenuation. An X-ray beam is restricted by collimators into a narrow fan or cone shape. As this beam transverses the patient, its intensity is attenuated through photoelectric absorption and Compton scattering. The transmitted beam intensity is measured by an array of detectors situated directly opposite the X-ray tube. The data set representing the attenuation profile from a single angle is termed a "projection." The gantry rotates seamlessly, establishing millions of discrete measurements over 360°. A computer system applies complex mathematical algorithms—most notably, Filtered Back Projection (FBP) or modern Iterative Reconstruction (IR) techniques—to map these attenuation measurements into a matrix of pixels. Each pixel is assigned a CT number (Hounsfield Unit, HU) indicative of the linear attenuation coefficient of the corresponding tissue voxel. This rigorous mathematical foundation not only eliminates structural superimposition but dramatically empowers the detection of subtle low-contrast differences among soft tissues compared to standard radiography.

3.3 Description of the CT Machine Components

A diagnostic state-of-the-art CT machine is an amalgamation of intricate mechanical, electrical, hardware, and software subsystems functioning in extraordinary synchrony. The core components include the Gantry, Examination Table, the Computer System, and the Control Console.

3.3.1 The Scanning Gantry

The gantry is the dominant mechanical structure in the CT suite and houses the principal X-ray generation and detection assemblies. It typically features a central aperture (commonly 70 to 80 cm in diameter) through which the patient couch freely translates. To accommodate different anatomical orientations, the entire gantry structure can often be tilted (usually up to $\pm 30^\circ$) from the vertical axis.

- **X-ray Tube:** Modern CT tubes are engineered to sustain immense thermal loads resulting from continuous, high-mA scanning required in multislice or helical CTs. They utilize rotating tungsten-rhenium anodes with large anode heat capacities and high cooling rates. The focal spot size directly bounds the system's geometric resolution; hence, many tubes incorporate "flying focal spot" technology, alternating the electron beam position to double the sampling frequency and reduce aliasing artifacts.
- **Detector Array:** Operating directly opposite the X-ray tube, the detector array captures exiting attenuation photon flux and transduces it into proportional electronic signals. Contemporary multi-detector CT (MDCT) units utilize solid-state scintillation detectors (such as Gadolinium Oxysulfide, $\text{Gd}_2\text{O}_2\text{S}$, or specialized ceramics). These rare-earth scintillators absorb X-rays efficiently to produce visible light flashes, which are instantaneously registered by integrated photodiodes. Essential characteristics of CT detectors include high geometrical efficiency, superior capture efficiency, negligible afterglow, and high dynamic range.
- **Data Acquisition System (DAS):** Positioned intimately within the gantry alongside the detectors, the DAS is responsible for rapidly measuring the minute analog photoelectric signals emanating from the detectors. These weak currents are amplified and converted mathematically into discrete digital data streams via analog-to-digital converters (ADCs). The colossal data packages are subsequently relayed to the computer system.
- **Slip Ring Technology:** This revolution enabled continuous helical (spiral) scanning. Unlike early generations where cables tethered the rotating components—necessitating a scan-and-stop operation to unwind—slip rings utilize conductive concentric tracks and metallic brushes. This mechanism transmits electrical power to the rotating tube and smoothly transfers continuous streams of digital data back to the stationary computer framework, facilitating unceasing sub-second rotations.
- **Collimators:** Beam collimation dictates the dose profile and z-axis slice width. *Pre-patient collimators* shape the primary X-ray fan/cone, controlling patient dose by eliminating penumbral, divergent edges, and establishing the dose profile width ($N \cdot w$). *Post-patient collimators* are situated precisely over the detector faces, significantly minimizing the detection of scattered photon trajectories which would drastically degrade image contrast.

3.3.2 The Examination Table (Couch)

The patient table provides immense structural rigidity combined with remarkable positional precision. During helical acquisitions, the table uniformly feeds the patient into the aperture simultaneously with continuous X-ray tube rotation. The bed must be fabricated from low Z-material (often carbon fiber) to avert significant X-ray attenuation and beam-hardening artifacts where the beam crosses through the structure. It must maintain mechanical indexing accuracy up to fraction-of-a-millimeter precision beneath weight loads exceeding 200 kg.

3.3.3 The Computer and Reconstruction System

The high-performance computer infrastructure bridges raw output and displayable image. It resolves millions of simultaneous linear equations generated by the DAS in essentially real time. It governs the entire hardware, stores retrospective raw data, interfaces via DICOM to the PACS (Picture Archiving and Communication System), and enables 3-dimensional multi-planar rendering, fluoroscopic CT, and complex dosimetric records.

3.3.4 The Control Console

The control console acts as the solitary user interface situated safely behind the lead-glass partition. It consists of multiple monitors, dedicated keypads for parameter manipulation (e.g., kV, mA, slice

thickness, pitch), and communication systems coupling the operator visually and audibly to the patient via intercoms.

3.4 Facility Design and CT Room Layout (AERB Guidelines)

Since computed tomography utilizes high-energy X-rays and large cumulative doses, structural radiation shielding and logical facility architecture are imperative for protecting staff, public, and adjacent zones against secondary scattered rays. In India, the Atomic Energy Regulatory Board (AERB) stringently regulates these layouts.

The basic guidelines explicitly prescribe:

- **Minimum Room Size:** The room housing the CT gantry must be sufficiently large to maneuver patients completely (minimum 25 sq. m usually depending on unit dimensions; minimizing dimensions around 4m). This safely dilutes in-room scatter by the inverse-square law.
- **Shielding Integrity:** The primary mechanism of exposure to individuals outside the CT room is scattered radiation from the patient's body and the gantry itself. Walls typically must consist of robust masonry (e.g., 9-inch solid brick) or be internally lined with specified lead equivalence (frequently 2 mm Pb). The access doors strictly require equivalent wood-and-lead sandwich protection (2 mm lead equivalence) to ensure zero weak points.
- **Control Console Placement and Viewing Glass:** To prevent occupational exposure, the technologist must operate the scanner entirely from a distinct control room or a heavily shielded alcove. Direct visual observation is uncompromisingly established using a specialized lead-glass window. As per the AERB Model Layout in the appended document, the viewing window is specified as 100 cm × 80 cm maintaining a strict **2.0 mm lead equivalence**.
- **Operational Layout:** The layout must minimize the scatter angle. In the prescribed structural blueprint (refer to Figure 1), the Examination Table (1) moves perpendicularly relative to the structural geometry, while the Control Unit (3) is protected dynamically from immediate sideways scattering. Warning fixtures, prominently a bright *Red Light* and radiation warning placards strictly adorn the external facade to deter entry during beam-on periods.

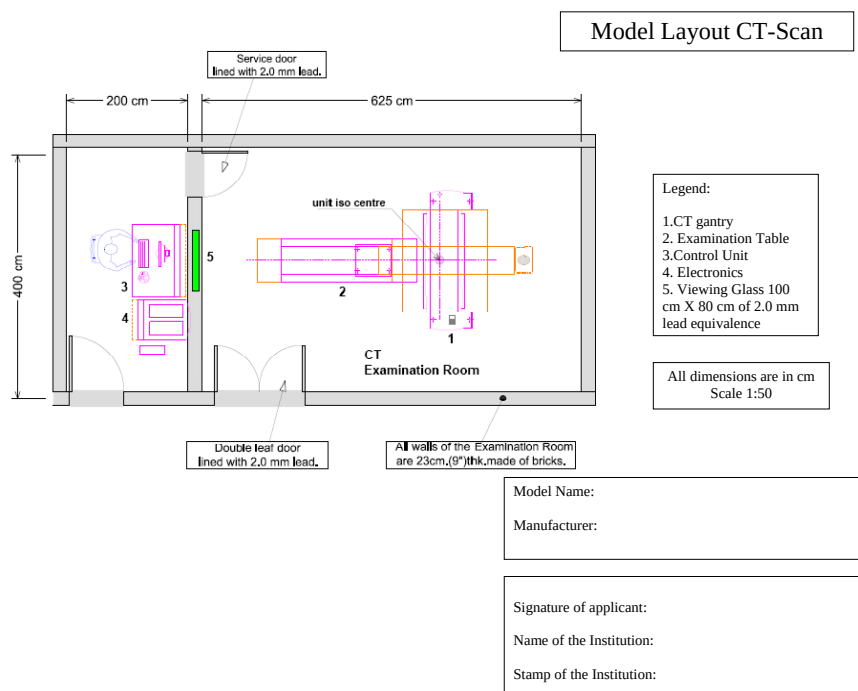


Figure 1: Model Room Layout for CT Scan Installation as prescribed by AERB. Components include: 1. CT gantry, 2. Examination Table, 3. Control Unit, 4. Electronics, 5. Viewing Glass (100 cm × 80 cm of 2.0 mm lead equivalence).

3.5 Quality Assurance (QA) Parameters

As per the guidelines mandated by regulatory bodies such as the Atomic Energy Regulatory Board (AERB), comprehensive acceptance testing upon new equipment installation and periodic QA are strict requirements. The QA tests for CT equipment are broadly classified into four primary categories: Mechanical Tests, Tests for High-Frequency Generators, Image Quality Parameters, and Radiation Safety [3].

3.5.1 Mechanical and Geometric Tests

The geometric fidelity of the scanner is critical to avoid image artifacts and ensure precise localization.

- **Alignment of the Table to Gantry:** This test ensures that the longitudinal axis of the patient table is perfectly horizontal and aligned with the vertical line passing through the scanner's rotational axis. A mismatch introduces severe image artifacts, particularly for bulky patients. The acceptable tolerance is ± 5 mm of the table midline.
- **Gantry Tilt Accuracy:** Most modern CT scanners allow non-orthogonal scanning by tilting the gantry. The accuracy of the displayed tilt angle is assessed by supporting an envelope-wrapped X-ray film vertically at the gantry end of the couch. The tolerance is typically $\pm 2^\circ$ [3].
- **Table Indexing Accuracy and Increment:** Couch translation must be precisely synchronized with the X-ray tube rotation. Using a ruler or tape measure placed alongside the table, the indicated degree of couch movement is compared with the actual physical distance moved. Tolerances specify differences extending over ± 1 mm.
- **Laser Positioning Accuracy:** Both internal gantry lasers and external wall-mounted alignment lasers guarantee iso-centering the patient. Max acceptable limit is typically ± 2 mm.

3.5.2 Tests for High-Frequency Generators

The X-ray generator parameters directly dictate the photon fluence and beam quality, directly impacting patient dose and image contrast.

- **Accuracy of Operating Potential (kVp):** The applied kilovoltage affects both the quality and quantity of the X-rays. Variations from the prescribed kVp will negatively alter image contrast. Performance bounds fall within $\pm 5\%$.
- **Reproducibility of Radiation Output:** Keeping mA and time fixed, the output is measured repeatedly at available kVp stations. Consistency is evaluated via the Coefficient of Variation (COV), maintaining $\text{COV} \leq 0.05$.
- **Linearity of Radiation Output (mA/mAs Linearity):** Keeping the kVp and exposure time constant, radiation output is measured across various mA stations. The Coefficient of Linearity (COL) is calculated spanning contiguous values and must fall under ≤ 0.1 .
- **Half-Value Layer (HVL) / Total Filtration:** Added filtration removes low-energy X-ray photons that do not possess sufficient energy to penetrate the patient and reach the detectors.

3.5.3 Image Quality Evaluation

Image quality parameters are assessed using standardized test phantoms of human shape or specific uniform test objects provided by the manufacturer.

- **CT Number Accuracy and Uniformity:** CT numbers (Hounsfield Units) depend on tube voltage, beam filtration, and object attenuation. By definition, the CT number of water is 0.
 - *Accuracy:* The measured CT number in the central Region of Interest (ROI) of a water phantom must be 0 ± 4 HU.
 - *Uniformity:* The CT number of a homogeneous object should remain constant across the entire field of view. The deviation between any peripheral ROI and the central ROI must not exceed ± 4 HU.

- **High Contrast Resolution (Spatial Resolution):** Defines the minimum size of a detail visualized with a contrast $> 10\%$. Expressed as spatial frequency (line pairs/cm), aiming normally for > 15 lp/cm.
- **Low Contrast Resolution:** This metric represents the smallest visible object at a specific low contrast level. It determines the system's capacity to differentiate tissues with minimal density differences relative to the surrounding background.
- **Slice Thickness (Z-axis Resolution):** Assesses whether the effective physical slice thickness matches the nominally selected thickness on the console. Tolerances define an error bracket ≤ 1 mm for sub-millimeter slices and $\pm 10\%$ for > 2 mm.

3.5.4 CT Dosimetry and Radiation Safety

The primary objective of these metrics aims exactly mapping standard absorbed doses.

- **Computed Tomography Dose Index ($CTDI_{100}$):** Incorporates practical parameters using a standard 100 mm pencil ionization chamber internally fixed inside standard body phantoms registering core dose lengths mathematically via:

$$CTDI_{100} = \frac{1}{N \cdot w} \int_{-50}^{50} D(z) dz \quad (1)$$

- **Weighted CTDI ($CTDI_w$):** The weighted CTDI is defined as

$$CTDI_w = \frac{1}{3}CTDI_{100,center} + \frac{2}{3}CTDI_{100,periphery} \quad (2)$$

where the peripheral value is the average of the four measurements taken at the periphery of the CT phantom:

$$CTDI_{100,periphery} = \frac{1}{4}(CTDI_{12 \text{ o'clock}} + CTDI_{3 \text{ o'clock}} + CTDI_{6 \text{ o'clock}} + CTDI_{9 \text{ o'clock}}) \quad (3)$$

- **Volume CTDI ($CTDI_{vol}$):** For helical CT, the volume CTDI is obtained by correcting for table movement using pitch:

$$CTDI_{vol} = \frac{CTDI_w}{pitch} \quad (4)$$

- **Dose Length Product (DLP):** Another useful output quantity is the dose-length product (DLP), which represents the total scanner output over the scanned length:

$$DLP = CTDI_{vol} \times \text{scan length} \quad (5)$$

Thus, CT quality assurance involves verification of image quality, geometric accuracy, CT number calibration, and dose indices. These checks ensure that the scanner produces diagnostically reliable images with acceptable radiation output and consistent performance over time.

3.6 Quality Assurance Equipment and Instrumentation

A comprehensive quality assurance protocol demands specialized, highly accurate instrumentation to measure both dosimetric output and image quality fidelity. The following key apparatuses are standard in modern CT QA procedures.

3.6.1 GE Optima CT580W CT Scanner

The GE Optima CT580W is a large-bore (80 cm), high-performance 16-slice computed tomography scanner. It is predominantly designed to accommodate patients requiring greater clearance (such as trauma or bariatric cases) and is widely used as a simulation scanner in radiation oncology due to its geometry allowing the inclusion of immobilization devices. Evaluating the geometric and output parameters of this scanner is crucial because any inaccuracy directly influences treatment planning and spatial localization [4].



Figure 2: RTI Piranha Multi-Meter for comprehensive X-ray quality assurance measurements.

3.6.2 RTI Piranha Multi-Meter

The RTI Piranha is a comprehensive, multi-modal X-ray quality assurance meter designed to assess multiple fundamental physical parameters of the X-ray tube in a single exposure. Connected directly to its proprietary software (Ocean Next), it can instantly capture, analyze, and store data regarding the operating kilovoltage (kVp), exposure time (ms), Half-Value Layer (HVL), total filtration, dose, and dose rate. Its high accuracy in varying radiation fields makes it an essential tool for verifying generator performance and establishing baseline CT output linearity and reproducibility [5].

3.6.3 RTI CT Ionization Chamber (10 cm)

The measurement of Computed Tomography Dose Index ($CTDI_{100}$) strictly requires a specialized pencil-type chamber. The RTI CT Ionization Chamber is a robust cylindrical detector with a precise 100 mm active detector length, constructed to fit seamlessly into the conventional acrylic head (16 cm diameter) and body (32 cm diameter) dosimetry phantoms. When connected to an electrometer (such as the RTI Piranha or Cobia), the chamber accurately intercepts the scattered and primary X-rays across overlapping slices, enabling accurate dosimetric integral mapping and calculation of $CTDI_w$ and $CTDI_{vol}$ [6].



Figure 3: Key quality assurance equipment: RTI CT Ionization Chamber

3.6.4 Pro-CT MINI Phantom

The Pro-CT MINI is a versatile, compact physical phantom meticulously structured for routine and extensive image quality control tests. Constructed homogeneously from a water-equivalent tissue-mimicking



Figure 4: Pro-CT MINI Phantom for comprehensive image quality testing.

material, it features internal structural modules dedicated to testing various fundamental imaging parameters within a single scan. It prominently facilitates the calculation of CT Number (Hounsfield Unit) accuracy for water-equivalence, image uniformity across peripheral and central Regions of Interest (ROI), high-contrast spatial resolution (via integrated line-pair patterns), and low-contrast detectability. The phantom ensures the CT reconstruction algorithms are constantly producing authentic diagnostic details free from artifacts [7].

4 Observation

4.1 Mechanical Tests

4.1.1 Gantry Tilt: Accuracy of the tilt Indicator

Most CT scanners accomplish non-orthogonal scans by tilting the gantry with some angle. If the tilt is not accurate, the gantry may hit the patient for some extreme clinical conditions. For the measurement radiochromic film was used.

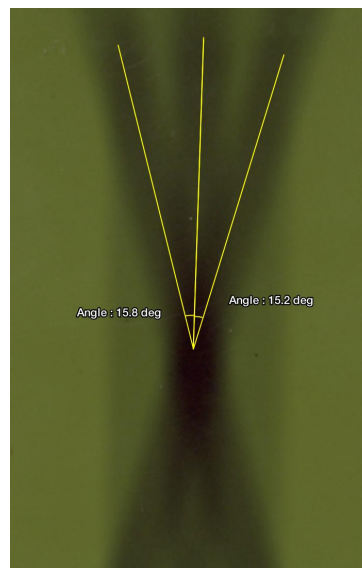


Figure 5: Gantry Tilt Measurement using Radiochromic Film

Actual gantry tilt = ± 15 degrees Measured gantry tilt = 15.8, 15.2 degrees Difference = 0.8, 0.2 degrees Tolerance: The measured gantry tilt should be within ± 2 degree of the actual gantry tilt. So, the measured gantry tilt is within the tolerance limit of ± 2 degree.

4.2 Radiation Safety Test of Computed Tomography Equipment

4.2.1 Slice Thickness

Slice thickness in the Pro-CT Mini is determined by measuring the length of ramps in the geometrical module image. The measured length depends on the slice thickness and an angle at which the ramp is inclined.

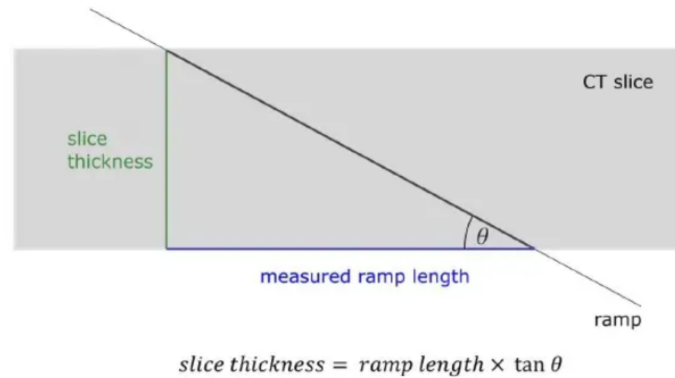


Figure 6: Geometry used in the Pro-CT Mini phantom for measurement of slice thickness

In the pro-CT mini phantom, ramps are at 45 degree angle so $\tan(45^\circ) = 1$, hence the measured length of the ramp is equal to the slice thickness. It is taken from this [ref.](#)

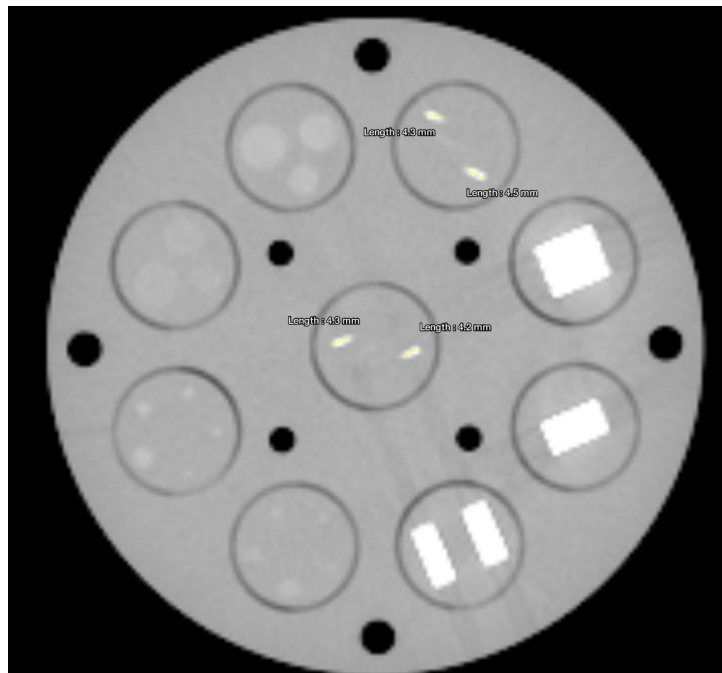


Figure 7: Slice Thickness Measurement using Pro-CT Mini Phantom

Set slice thickness = 5 mm
Measured slice thickness = 4.3 mm, 4.2 mm, 4.3 mm, 4.5 mm
Average measured slice thickness = 4.325 mm

AERB Tolerance:

For less than 1 mm : 0.5

1 mm to 2 mm : $\pm 50\%$

Above 2 mm : ± 1 mm

So it is under the tolerance limit of ± 1 mm.

4.2.2 Accuracy of Operating Potential

KV	50 mA	200 mA	400 mA	Avr	Diff	Tol	Remarks
80.0	79.0	78.7	78.1	78.6	1.4	± 2 kVp	within tol
100.0	98.2	97.9	97.5	97.9	2.1		out of tol
120.0	118.2	117.7	117.5	117.8	2.2		out of tol
140.0	137.9	137.5	137.6	137.7	2.3		out of tol

4.2.3 Accuracy of Timer

Set Time(ms)	Observed time(ms)			Avr time	% diff	Tol	Remark
500	508.4	508.4	507.9	508.23	1.64667	± 10 %	within tol
1000	1008	1009	1008	1008.33	0.83333		within tol
1500	1508	1508	1509	1508.33	0.55556		within tol

4.2.4 Linearity of Radiation Output

Operating potential = 120 kVp

mAs	Output (mGy)			Average Output (mGy)	mGy/mAs(X)	CoL
50	4.561	4.562	4.561	4.56	0.0912	0.00829
200	18.11	18.1	18.1	18.1	0.0905	
400	36.03	36.04	36.03	36.03	0.0901	
666	59.69	59.76	59.77	59.74	0.0897	

4.2.5 Reproducibility of Radiation Output

Operating current = 100 mA

kVp	O/P(mGy)				mean	COV
80	3.877	3.875	3.876	3.875	3.87575	0.00024703
100	6.959	6.951	6.959	6.958	6.95675	0.00055517
120	10.63	10.63	10.63	10.62	10.6275	0.00047048

4.2.6 Radiation Dose Test Measurement of Weighted CT Dose Index ($CTDI_w$)

Operating potential = 120 kVp Tickness = 0.5 cm Operating current = 200 mA

CT phantom type	Position	CT exposure (mGy cm)	CTDI(mGy)	CTDI _w
Head	A(Center)	17.16	34.31	37.44
	B(12 O' clock)	20.74	41.49	
	C(3 O' clock)	19.58	39.17	
	D(6 O'Clock)	17.51	35.03	
	E(9 O' clock)	20.17	40.33	
Body	A(Center)	5.261	10.52	16.885
	B(12 O' clock)	10.42	20.85	
	C(3 O' clock)	9.37	18.74	
	D(6 O'Clock)	8.543	17.09	
	E(9 O' clock)	11.79	23.59	

Tolerance: ± 20 % of the quoted value for head phantom and ± 40 % of the quoted value for body phantom.

4.2.7 Low Contrast Resolution

Phantom = Pro-CT Mini Phantom

window width = 395, window level = 210

Low Contrast Resolution = 2 mm at 0.5 % contrast level.

Tolerance: 2.5 mm at 0.5 % contrast difference (expected) or 5 mm at 1 % contrast difference (minimum).

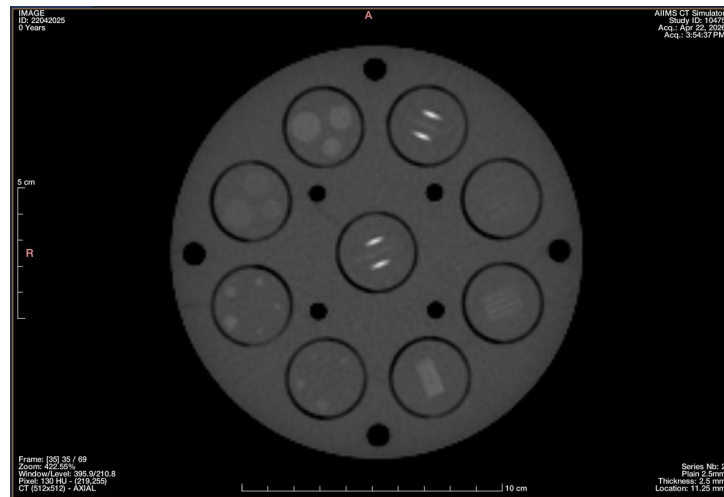


Figure 8: Low Contrast Resolution Measurement using Pro-CT Mini Phantom

The low contrast resolution is 2 mm at 0.5 % contrast level which is within the tolerance limit of 2.5 mm at 0.5 % contrast difference, hence the low contrast resolution of the CT machine is acceptable.

4.2.8 High Contrast Resolution

Phantom = Pro-CT Mini Phantom

window width = 490, window level = 307

Size of the smallest resolvable bar pattern = 5 lp/cm.

Tolerance: At 10 % contrast difference the size of the bar pattern that could be resolved should be 1.6 mm (3.12 lp/cm), expected high contrast resolution: 0.8 mm (≈ 6.25 lp/cm).

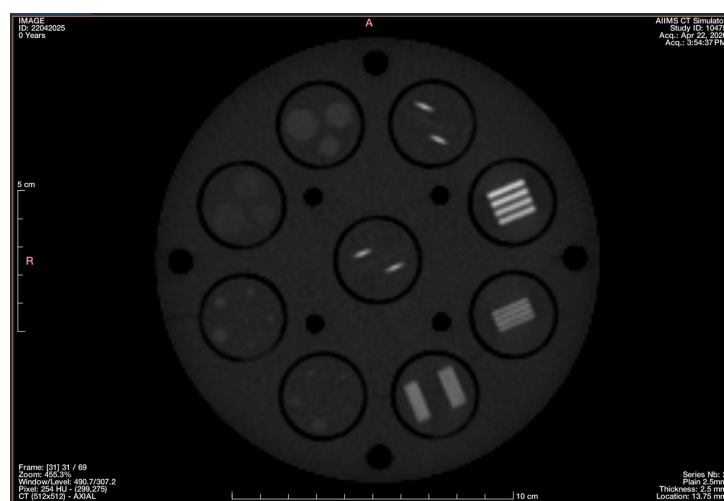


Figure 9: High Contrast Resolution Measurement using Pro-CT Mini Phantom. Max. resolved line pair pattern for measurement of high contrast resolution is 5 lp/cm.

4.2.9 Radiation Leakage Level from X-ray Tube Housing

Operating potential = 140 kVp Operating current = 500 mA time = 1 sec

Location	Radiation level ($\mu\text{Gy/hr}$)	Max leakage radiation ($\mu\text{Gy/hr}$)	Remark
Front (cathode)	1326	0.02213	within tol
Back(anode)	605.9		
Right	15.12		

4.2.10 Radiation Protection Survey of the Installation

Location	Radiation level ($\mu\text{R/hr}$)	Max Rad Level(mR/week)	Remark
Service door	104	0.0104	Within tol
operator	26	0.0026	Within tol
viewing window	70	0.007	Within tol
console door	19	0.0019	Within tol
Patient waiting area	11	0.0011	Within tol
Patient entry door	115	0.0115	Within tol

5 Conclusion

In this experiment, comprehensive Quality Assurance (QA) testing of a diagnostic computed tomography (CT) machine (GE Optima CT580W) was performed in accordance with regulatory guidelines to verify its mechanical accuracy, dosimetric output, image quality, and radiation safety. The evaluations were conducted utilizing specialized equipment: the Pro-CT MINI phantom, RTI Piranha multi-meter, and RTI CT Ionization Chamber.

The assessment of mechanical and geometric fidelity revealed that the gantry tilt indicator is highly accurate, showing a maximum deviation of only 0.8° (within the acceptable $\pm 2^\circ$ tolerance). The slice thickness verification corroborated these findings; for a nominal 5 mm slice, the average measured thickness was 4.325 mm, securely fulfilling the AERB tolerance requirement of ± 1 mm.

Output performance parameters—specifically the accuracy of the operating potential (kVp), timer accuracy, output linearity, and output reproducibility—demonstrated consistent and stable generator performance throughout the clinical range. Furthermore, the dosimetric assessment measuring the Weighted CT Dose Index ($CTDI_w$) confirmed that patient radiation exposures align predictably with the manufacturer's operational specifications and normative tolerance bounds.

Image quality was strictly evaluated using the Pro-CT MINI phantom:

- **Low Contrast Resolution:** The system successfully resolved low-contrast details of 2 mm at a 0.5% contrast difference, surpassing the 2.5 mm minimum expected tolerance.
- **High Contrast Resolution:** Spatial detail verification resolved line pair patterns up to 5 lp/cm, comprehensively satisfying the minimum clinical requirement (3.12 lp/cm).

Finally, radiation leakage mapping and the radiation protection survey around the installation showed conclusively that stray and transmitted radiation levels around the scanner boundaries and housing are safely restricted within established regulatory limits.

Overall, all evaluated physical and functional parameters of the diagnostic CT scanner fell safely within operational tolerances. The equipment functions reliably, efficiently maintains patient radiation safety protocols, and continues to produce images of high diagnostic fidelity.

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